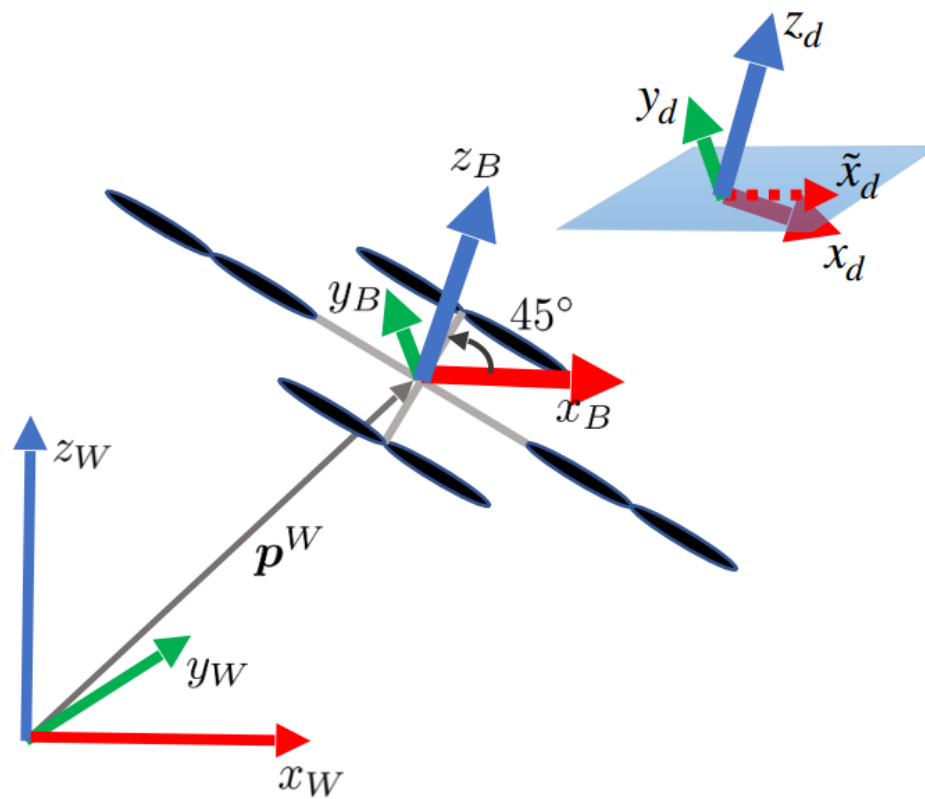


Guideline for Geometric Controller

VNAV2023

$$R_B^W = [x_B^W \ y_B^W \ z_B^W]$$

$$R_d^W = [x_d^W \ y_d^W \ z_d^W]$$



Steps for designing a geometric controller

- Input: desired position $P_d^w = [x_d, y_d, z_d]$ and desired yaw angle ψ_d
 - Output: propeller speed $\omega_i, i = 1, 2, 3, 4$
1. Compute $R_d^w = [x_d^w, y_d^w, z_d^w]$
 2. Compute $\tilde{x}_d^w$ which is the projection of x_d^w on x-y plane (in the world frame)

$$\tilde{x}_d^w = [\cos(\psi_d), \sin(\psi_d), 0]^T$$
 3. compute z_d^w using equation (7.13) in lecture notes
 1. Compute R_d^w using the equation right before 7.13 lecture notes and after 7.12
 2. Compute f_z^B and τ^B using equation (7.11) and 7.12
 3. Compute $\bar{F}$ from F in equation 6.9 by ignoring the first two rows
 4.
$$\begin{bmatrix} \omega_1 |\omega_1| \\ \omega_2 |\omega_2| \\ \omega_3 |\omega_3| \\ \omega_4 |\omega_4| \end{bmatrix} = \bar{F}^{-1} \begin{bmatrix} f_z^B \\ \tau^B \end{bmatrix}$$